Antenna bias: RC_BIAS, 33 ohm, 100nF to GND, LQG15HS27NJ02 onto RF_IN, ESD0P2RF at the U.FL. Choke is not in series with RF.

J3 X1200 harness: pin 1 to 5: +3V3, GND, SDA, SCL, GPIO26. R7 and R8 I2C pull-ups are DNP; V_USB is grounded.

GPS backup: ML414H direct on V_BCKP. Charge is LM4040-3.0, 34k, PMEG4005EJ. CR2032 feeds only RV-3028 VBACKUP.

ICM-45686 0x68, BMP581 0x47, RV-3028-C7 0x52, NEO-M9N UART+PPS, No UPS.

Kingfisher Sensor HAT rev 0.3

ML414H GPS backup, CR2032 RTC, X1200 harness, no UPS ICM-45686 + BMP581 + NEO-M9N + RV-3028	
westphae/kingfisher	
Sheet: /	
File: kingfisher-hat.kicad_sch	
Title: Kingfisher Sensor HAT	
Size: A1 Date: 2026-10-17	**Rev: 0.3**
KiCad E.D.A. 9.0.2	Id: 1/1